ALKEMY

AI PRODUCTIVITY

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THE PROBLEM

AI typically increases output and reduces time, but it isn’t quite obvious whether it improves business value.

OUR GOAL

We wanted to focus on where AI creates value, where it introduces loss, how it affects speed and quality, and whether there is a point where it becomes harmful.

METHOD

- **We cleaned the provided data to omit:**
  - Useless information (task_id, client, etc.)
  - Improperly labeled values
  - Missing/negative values (billable_hours, etc.)
- **We primarily analyzed data using:**
  - AI usage (percentage & boolean)
  - Profit
  - Rework
  - Outcome score
  - Task type
- **Our Approach:**
  - Compared AI vs. Non-AI tasks
  - Grouped data by AI usage levels
  - Calculated Percentage changes (speed, quality, rework)
  - Created heatmaps to compare key metrics (profit, outcome score, rework) across task types and AI usage
  - Identified patterns and thresholds





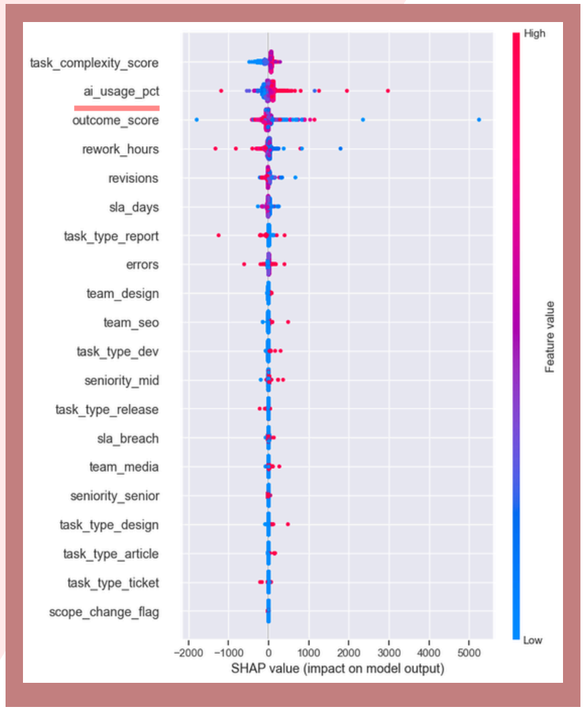




AI IMPORTANCE

We also used SHAP values with a gradient boosting model to identify the most important factors influencing each outcome. Across profit, time, errors, and quality, AI usage consistently appeared in the top four drivers. This shows that AI usage plays a central role in shaping performance across the system.

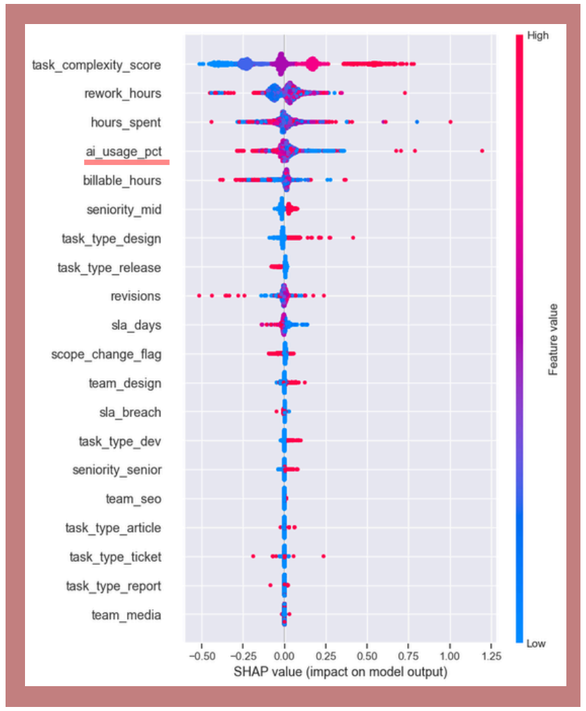
Profit



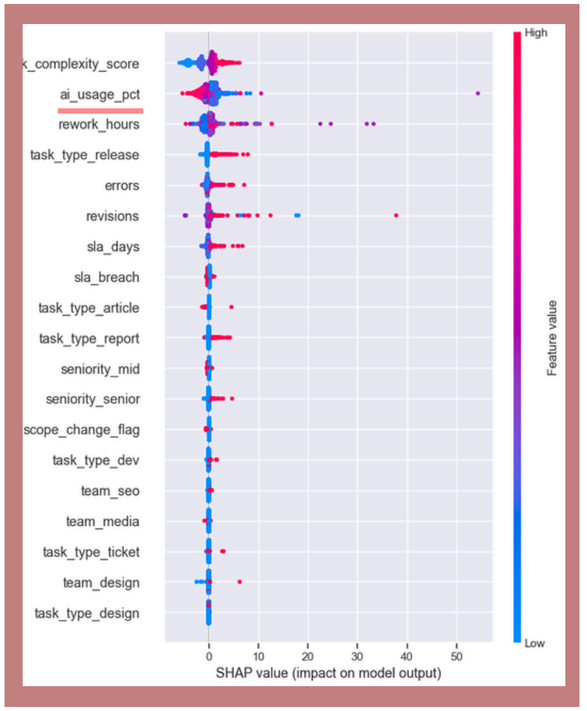
Outcome Score



Errors

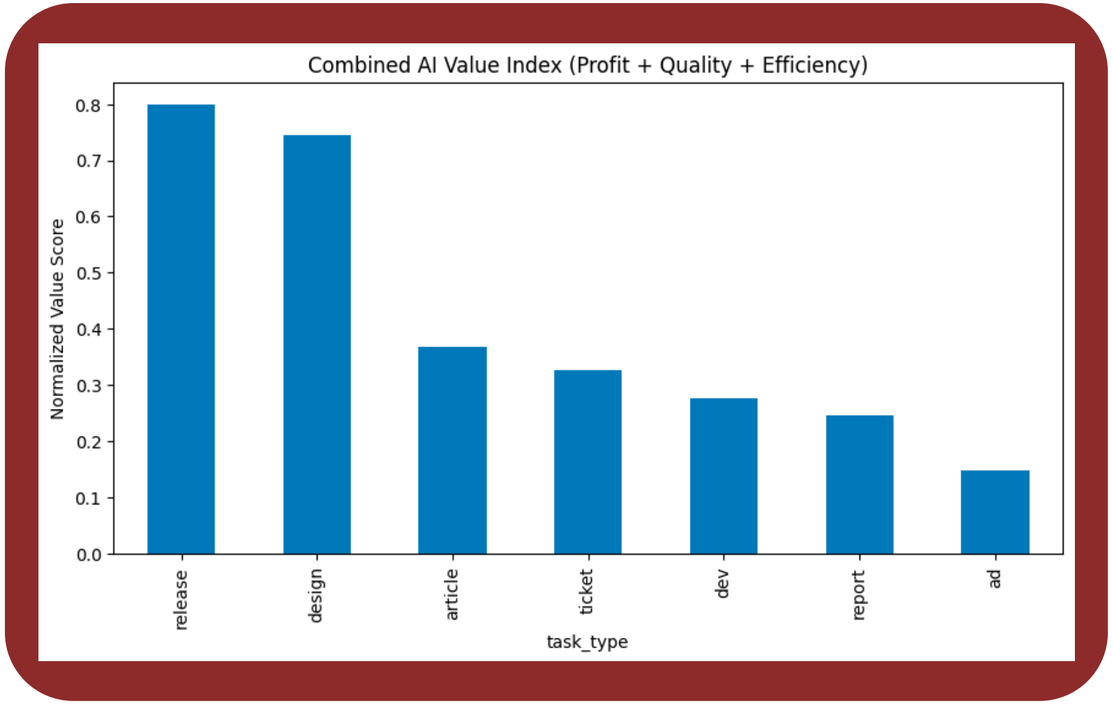


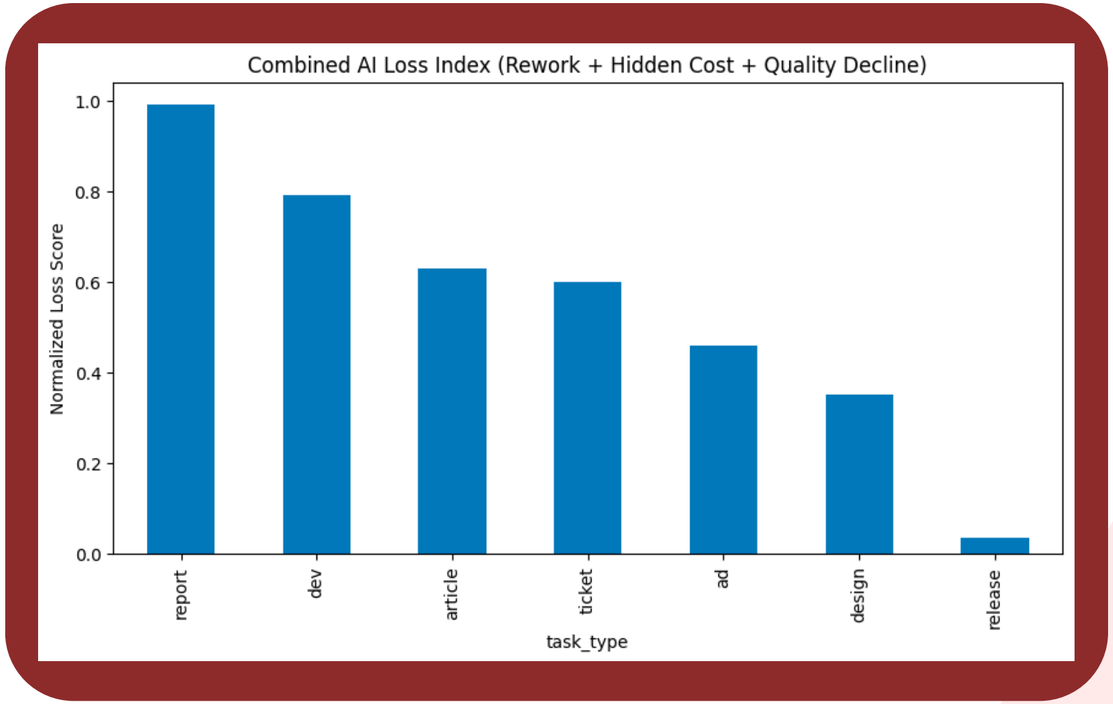
Hours Spent



INSIGHT 1

AI creates value, but not everywhere



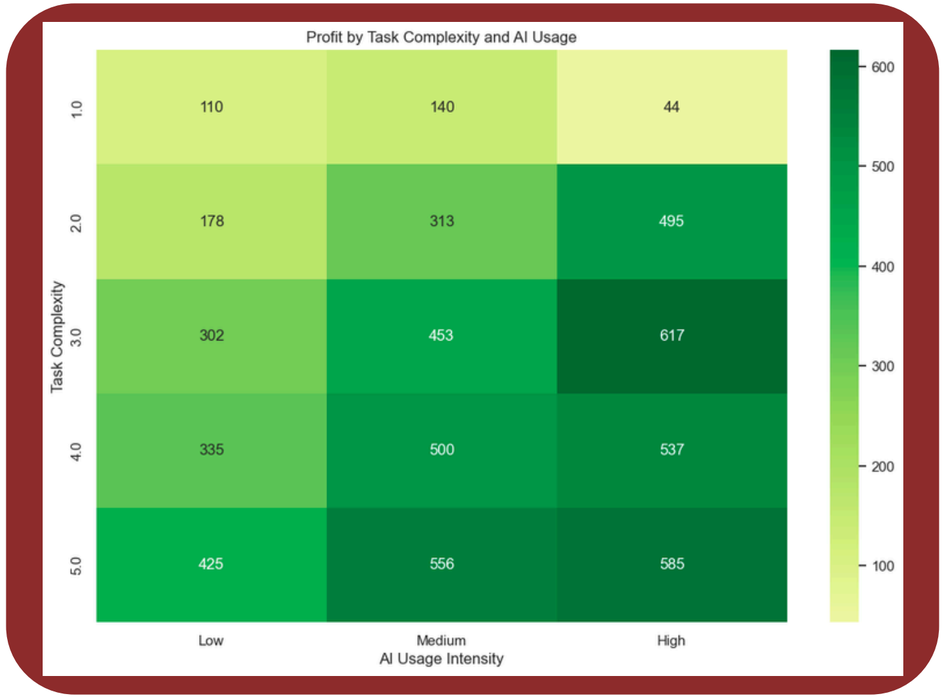


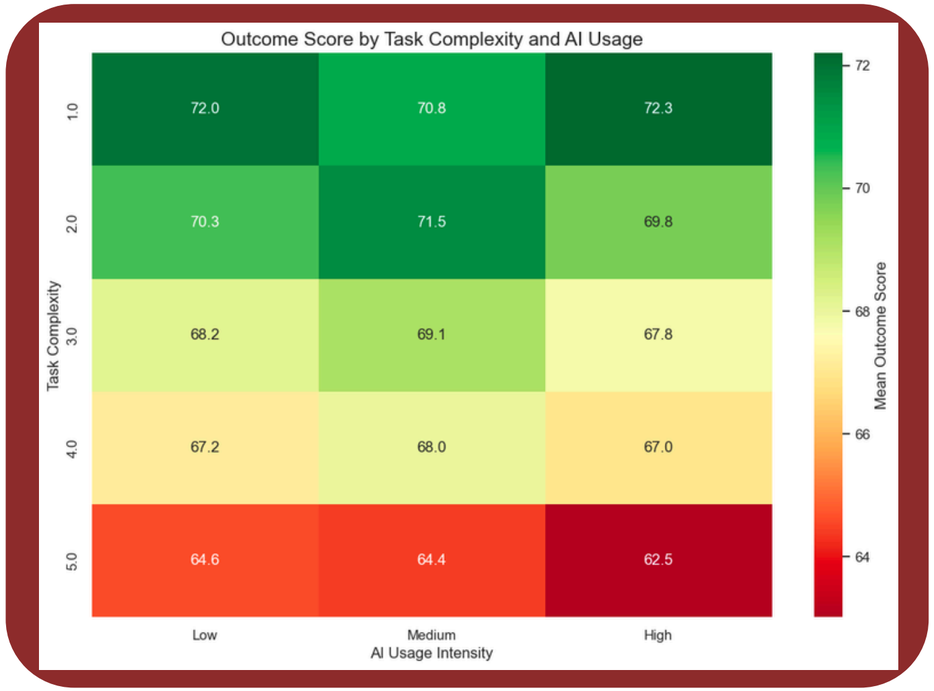
AI improves performance overall, but its impact varies by task, with the highest losses in complex tasks like reports and development. This variation is driven by both task complexity and task type.

INSIGHT 1A

AI impact depends on task complexity

- Higher AI usage leads to increased profit for most complexity levels.
- However, at the lowest complexity, moderate AI usage performs best.
- Outcome scores peak at moderate AI usage across most complexity levels.
- At the highest complexity, lower AI usage slightly outperforms.





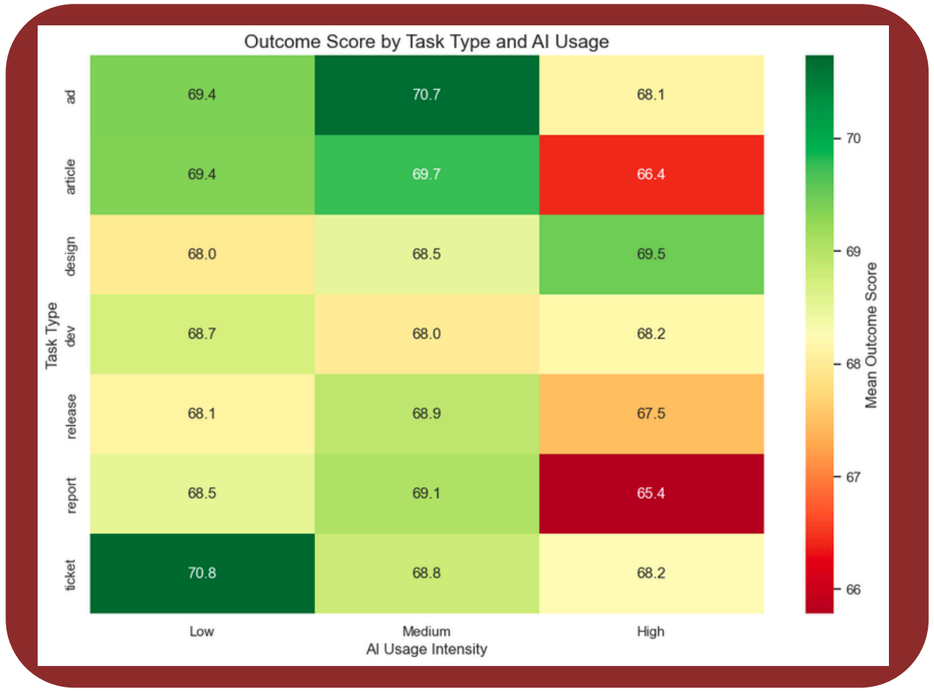
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INSIGHT 1B

AI effectiveness differs across task type

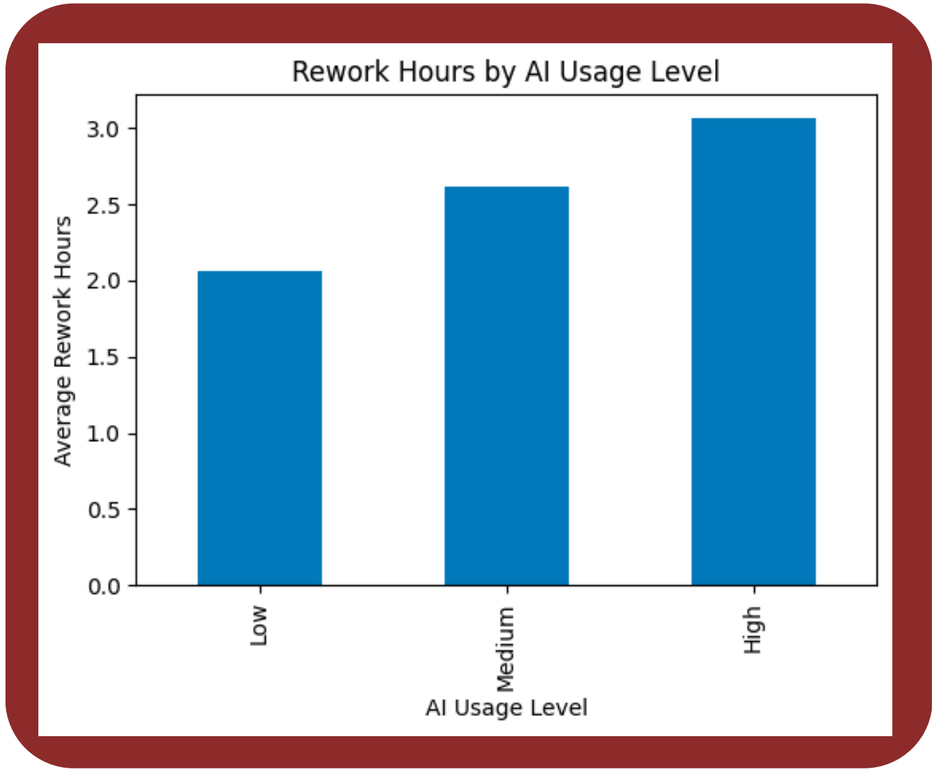
- High AI usage drives the greatest profits for structured tasks like release, report, and development.
- Moderate AI usage performs best for mixed tasks such as ads, articles, and design.
- Low AI usage is rarely optimal for profit.
- Outcome scores decline at high AI usage for most task types.
- Low to moderate AI usage produces the highest quality results.

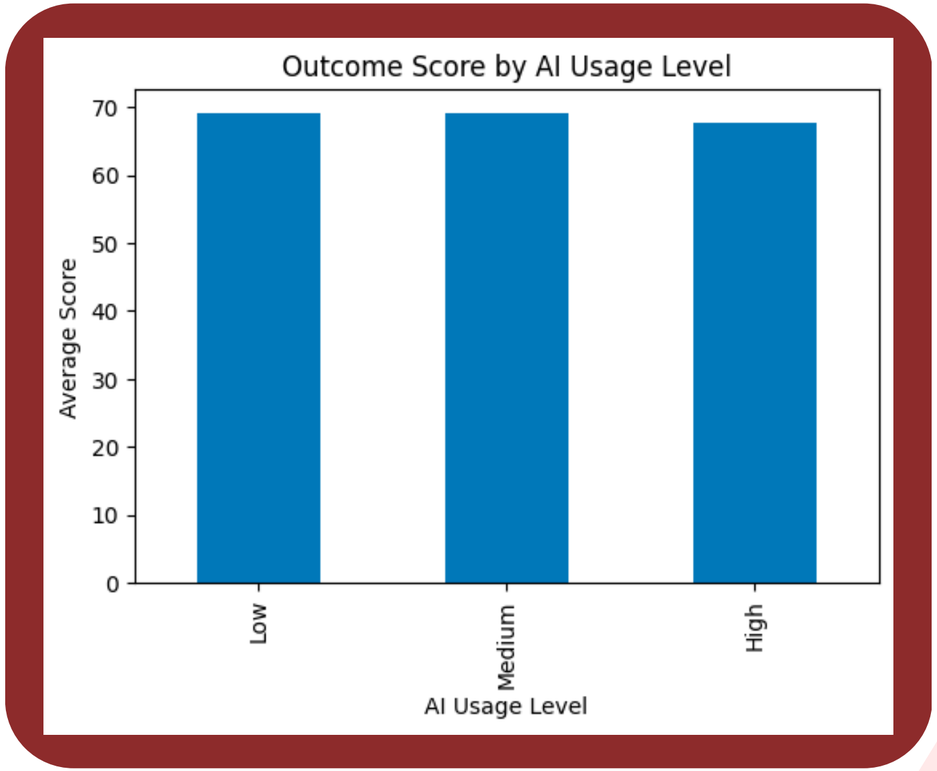




INSIGHT 2

Losses increase as AI usage rises due to rework and quality decline

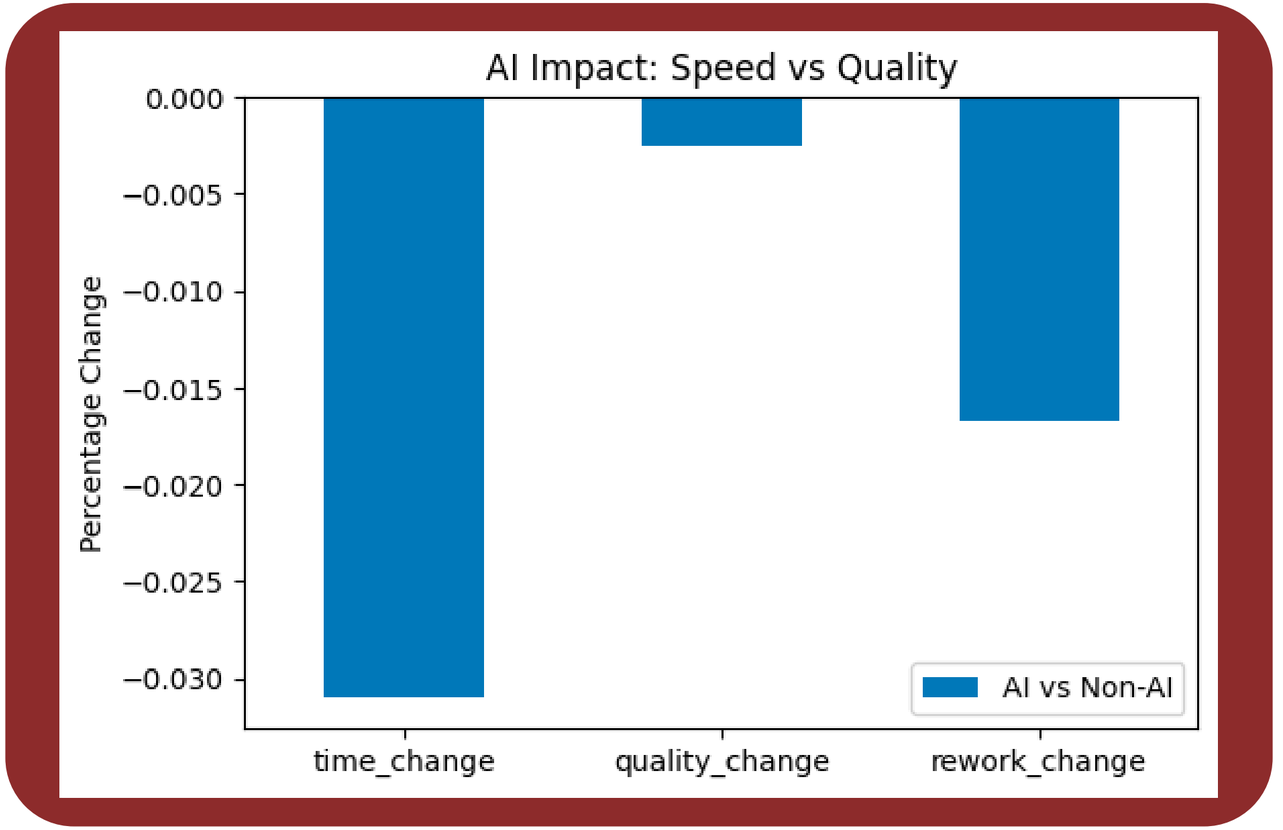




As AI usage increases, we see a clear rise in rework hours, meaning more time is spent fixing mistakes. At the same time, quality slightly decreases at higher levels of AI usage. This shows that losses are not coming from speed, but from errors and declining quality.

INSIGHT 3

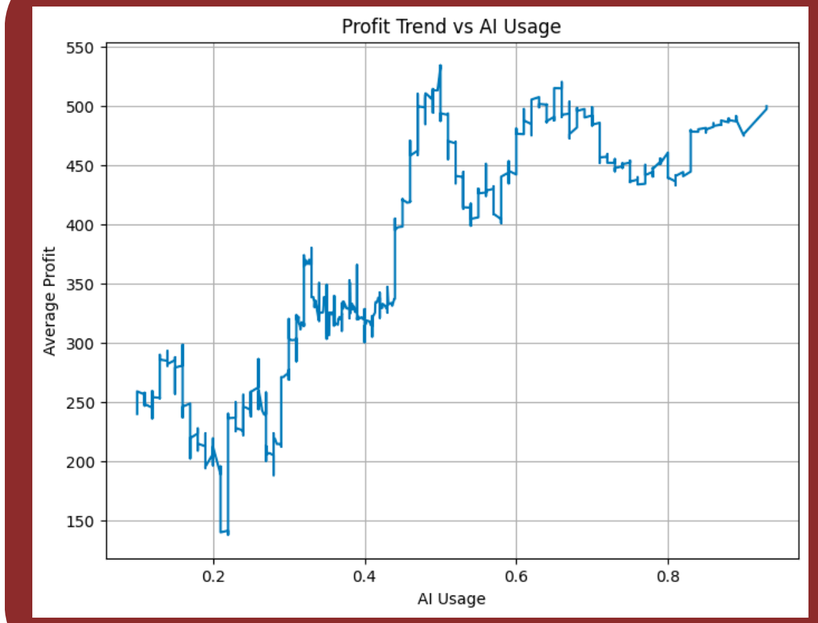
AI improves speed but slightly reduces quality

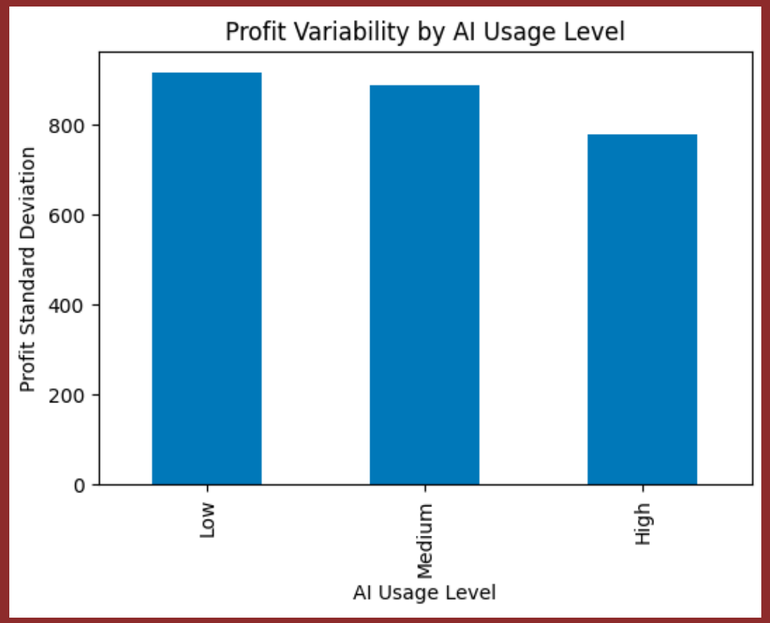


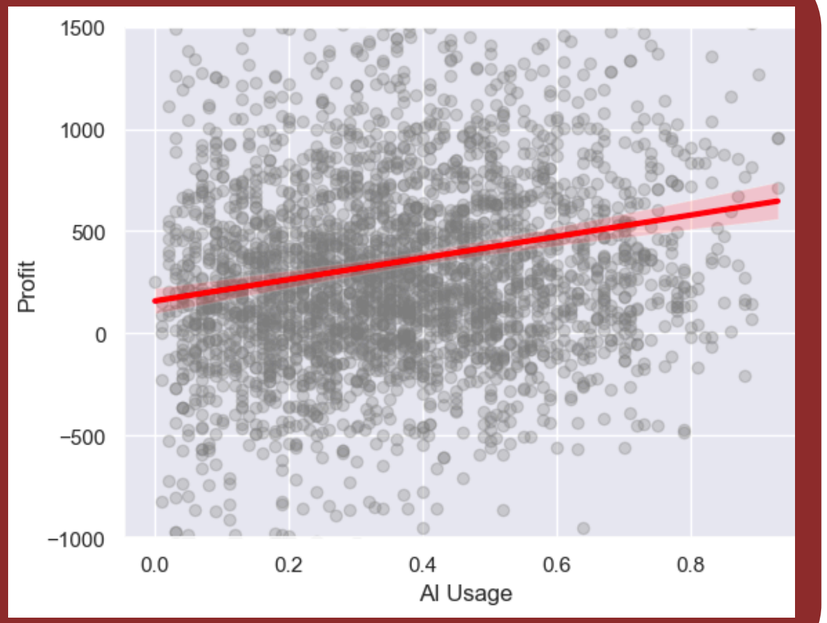
When we compare AI-assisted tasks to non-AI tasks, we see that AI reduces time but slightly lowers quality. This shows a trade-off between efficiency and accuracy.

THRESHOLD

Profit shows a noticeable increase beyond 45% AI usage. In addition, profit variability decreases at higher levels of AI usage, suggesting more consistent outcomes. However, taking into account how excessive AI usage increases rework, hidden inefficiencies are also revealed past this point. Taken together, this indicates that AI stops being fully beneficial at around 60% usage, where additional gains are offset by increasing rework which diminishes returns.

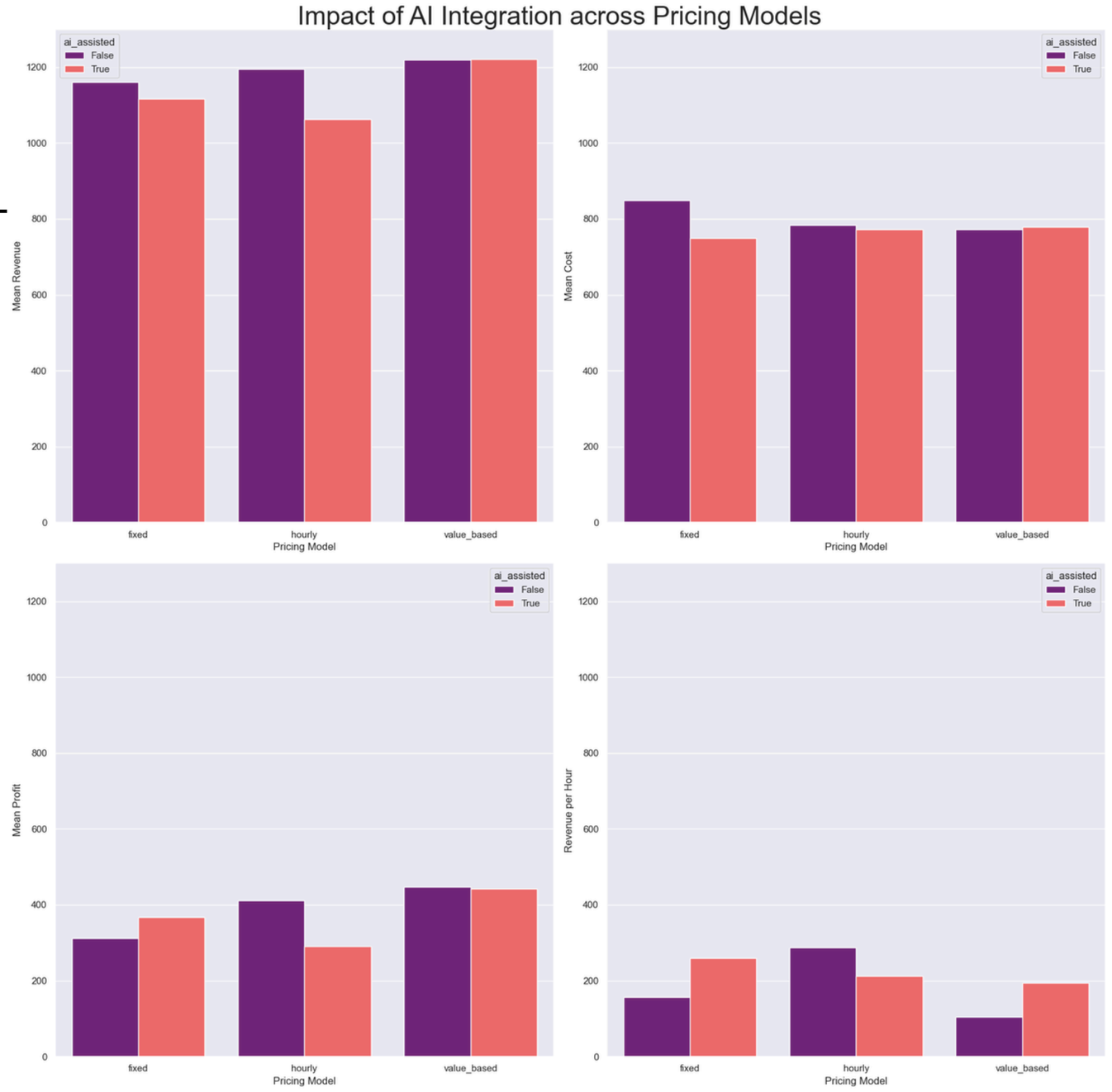






PRICING MODELS

- We compared pricing models using revenue, cost, profit, and revenue per hour with and without AI.
- The value-based model remains stable regardless of AI use.
- Fixed pricing benefits from AI through lower costs and higher profits.
- The hourly model shows lower profits and reduced revenue per hour with AI.
- This suggests that AI is least compatible with hourly pricing because efficiency gains are not captured in revenue.



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Business Decision

1. Shift away from hourly pricing
   * AI reduces billable hours and lowers revenue under hourly models.
   * Fixed and value-based pricing better capture efficiency gains.

2. Operate within the optimal AI usage range (40-60%)
   * Profit improves significantly beyond ~40% AI usage.
   * Above ~60%, gains are offset by rising rework.

3. Control AI usage by task type
   * High AI works best for structured tasks (release, design).
   * Moderate or low AI is better for complex or creative work.

4. Build quality control into workflows
   * AI increases rework and errors.
   * Add review checkpoints rather than relying on perfect AI outputs.

5. Track rework alongside profit
   * Profit alone is misleading.
   * Rising rework can quietly reduce long-term value.

WHAT DATA IS MISSING?

What Variables are Missing?

- Type of AI tool used (ChatGPT, Github CoPilot, etc.) and its purpose
- Experience with AI

What Metrics Should be Tracked?

- Client satisfaction or feedback
- Number of revisions per task
- Error rate of AI
- Long-term impact on quality

Questions we have for the Company?

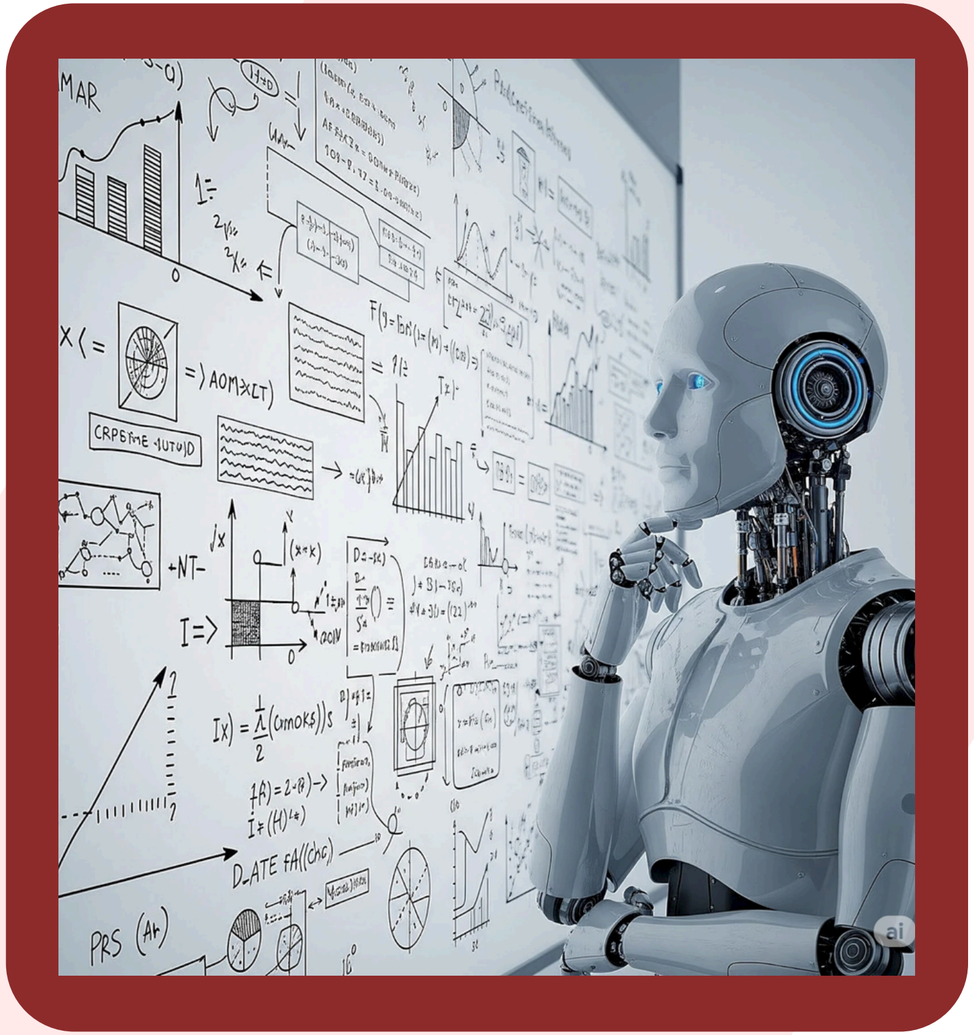
- Are there existing guidelines or limits on AI?
- Are there reoccurring errors with AI?
- How much time is spent reviewing AI output?

AI DISCOVERY

- AI can increase both efficiency and inefficiency at the same time.
- While higher AI usage leads to increased profit and faster task completion, it also introduces more rework and hidden costs.
- This shows that AI changes how value is created, often shifting effort from initial work to correction and review.

AI MISTAKE

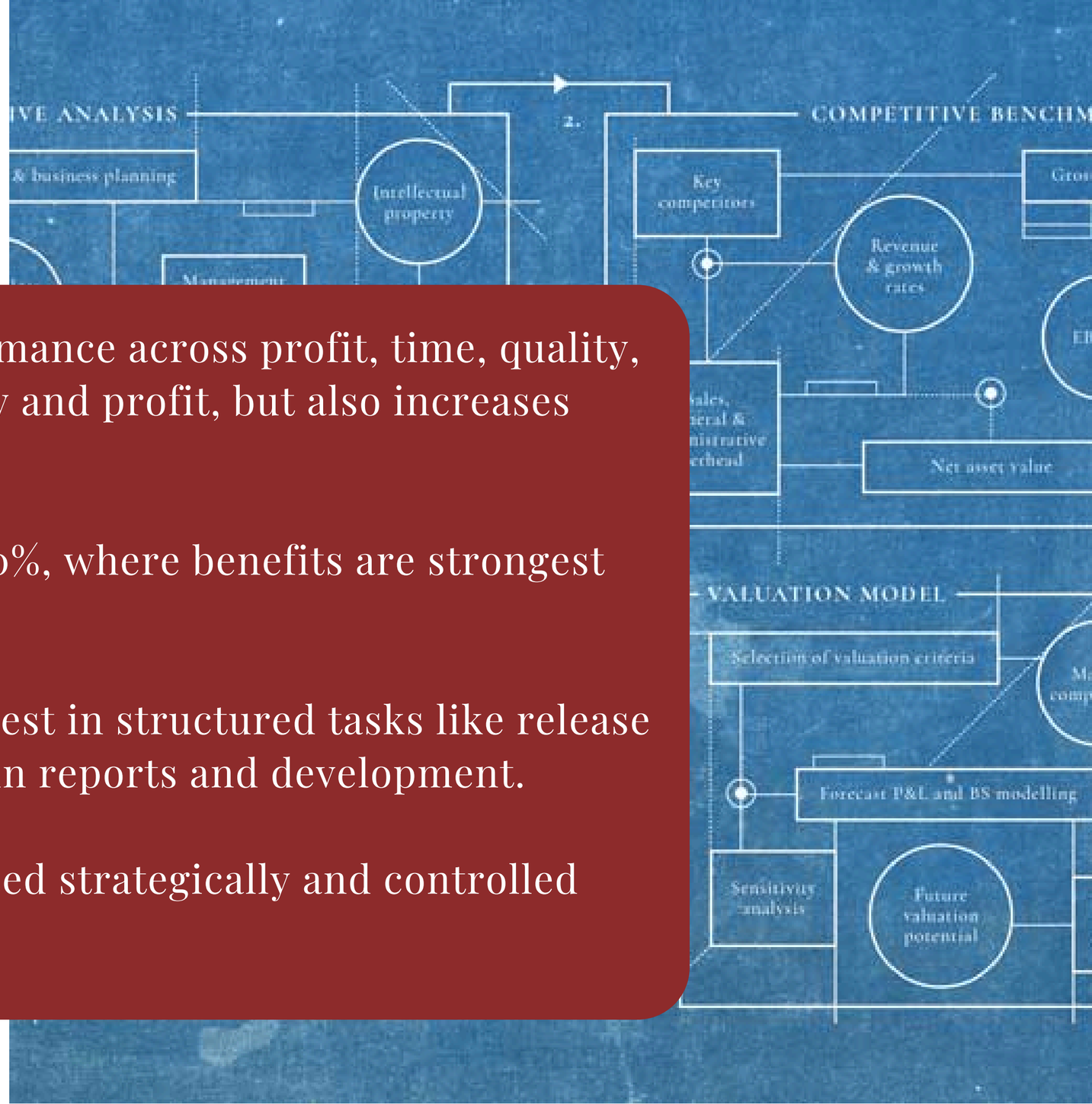
- We initally assumed that higher AI usage always leads to better performance.
- Early results suggested that increasing AI would continuously improve outcomes.
- Later we saw the hidden trade-offs such as increased rework and slight declines in quality.
- We can’t simply look at profit margins to determine the success rate of using AI.



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CONCLUSION

- AI usage is a key driver of performance across profit, time, quality, and errors. It improves efficiency and profit, but also increases rework and lowers quality.
- The optimal usage range is 40–60%, where benefits are strongest before trade-offs grow.
- Impact varies by task. AI works best in structured tasks like release and design, and performs worst in reports and development.
- Overall, AI creates value when used strategically and controlled carefully.





THANK YOU